

BernMathApplication

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1 Question 1

We take for granted that we're working in the reals, and not complex numbers. Call

$$f_a(x) = \frac{x^x - x^a}{a^x - a^a}$$

For the domain, since x^x appears in the function, we'll take $x \geq 0$, and won't analyze the cases where $x < 0$, since the function is not continuous. For the same reason, since a^a appears in the function, we'll consider $a \geq 0$.

The denominator is $a^x - a^a$. If $a = 1$, the denominator is null, and so the function $f_1(x)$ is simply undefined, there is no limit.

If $a = 0$, the term a^a is undefined. However, since $\lim_{a \rightarrow 0} a^a = 1$, we might want to take $0^0 = 1$ as a reasonable definition: in this case, $f_0(x) = 1 - x^x$, and so $\lim_{x \rightarrow 0} f_0(x) = 0$. In this case, we mean the limit as $x \rightarrow 0^+$, since we do not consider the limit for negative x .

If $a \neq 1, 0$, the function is correctly defined. Direct substitution of the limit gives a $0/0$ indeterminate form. We can use l'Hopital's rule, since both numerator and denominator are regular enough:

The derivative of the numerator, for $a > 0$, is

$$\partial_x(x^x - x^a) = x^x(\log x + 1) - ax^{a-1}$$

and the derivative of the denominator is

$$\partial_x(a^x - a^a) = \partial_x e^{x \log a} = a^x \log a$$

Substituting $x \rightarrow a$ in the limit, after applying l'Hopital, gives

$$\frac{a^a(\log a + 1) - aa^{a-1}}{a^x \log a} = \frac{a^a \log a + a^a - a^a}{a^x \log a} = 1$$

So the limit $\lim_{x \rightarrow a} f_a(x)$, when $a > 0, a \neq 1$ exists and is equal to 1 (left and right limit are thus equal).

2 Question 2

We take for granted the definition of a vector space V over a field K .

A set of vectors $v_1, \dots, v_n \in V$ is linearly independent if

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0_V \iff \alpha_1 = 0, \dots, \alpha_n = 0_K$$

where $\alpha_i \in K$, 0_V is the 0 vector in V , and 0_K is the additive identity element of the field K .

I first encountered this definition in the course of Linear Algebra 1, first year of university. From what I can remember, it did not remind me of any particular stuff I had seen before. While in highschool (thanks to Math Olympiads) I was already familiar with some university-level concepts, but they were mostly related to problem solving. Linear Algebra 1 is the first time I dealt with a mature theory where one can find these definitions.

3 Question 3

Let's take the partial derivative of $E[u]$ with respect to $u[k]$. Because of the linearity of the derivative operator:

$$\nabla_k E = \partial_{u[k]} E[u] = \partial_{u[k]} \sum_{i=1}^{N-1} (u[i] - u[i-1])^2 = \sum_{i=1}^{N-1} \partial_{u[k]} (u[i] - u[i-1])^2$$

We will use the Dirac delta notation: $\delta_{i,j}$ is defined to be 1 when $i = j$, 0 otherwise. The expression above is then

$$\sum_{i=1}^{N-1} \partial_{u[k]} (u[i] - u[i-1])^2 = \sum_{i=1}^{N-1} 2\delta_{k,i} (u[i] - u[i-1]) - 2\delta_{k,i-1} (u[i] - u[i-1])$$

Summing over i , the only terms that remain are the ones where the Dirac delta are not-null, so the above is equal to

$$2(u[k] - u[k-1]) - 2(u[k+1] - u[k]) = -2u[k+1] + 4u[k] - 2u[k-1]$$

for $k = 0, \dots, N-1$. Note that for $k = 0, N-1$ (the first and last indexes), the formula above has some terms that go beyond the permitted indexes (from 0 to $N-1$): in the original definition of $E[X]$, the first and last indexes only appear once, while every other index appear twice. The terms that have "beyond boundary" indexes thus do not contribute and are equal to 0.

The partial derivative $\nabla_k E = \partial_{u[k]} E[u]$ is then:

$$\nabla_k E[u] = -2u[k+1] + 4u[k] - 2u[k-1]$$

for $k = 1, \dots, N-2$. At the boundary indexes we instead have:

$$\nabla_0 E[u] = -2u[1] + 4u[0]$$

$$\nabla_{N-1} E[u] = 4u[N-1] - 2u[N-2]$$

I learned the formal definition of a gradient during Calculus 2. I heuristically understood it to be the "direction of steepest descent", i.e. the infinitesimal direction in the N-dimensional (real) space where the function $E[u]$ decreases the most.

When I switched from pure mathematics to machine learning, this definition made it easy to understand what we do when we minimize a loss function: at each step of the gradient descent, we "move" towards the direction that minimizes the (linear approximation of) the loss function.

4 Question 4 - Done, maybe explain better the steps

Let's compute the expectation $E[X]$. By definition of the expectation value and the PMF of the Poisson random variable:

$$E[X] = \sum_{x \geq 0} xp(x; \lambda) = \sum_{x \geq 0} x \frac{e^{-\lambda} \lambda^x}{x!}$$

The term for $x = 0$ is null, and we can take the term that does not depend on x :

$$\sum_{x \geq 0} x \frac{e^{-\lambda} \lambda^x}{x!} = \sum_{x \geq 1} x \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x \geq 1} \frac{\lambda^x}{(x-1)!}$$

We take out a λ factor:

$$e^{-\lambda} \sum_{x \geq 1} \frac{\lambda^x}{(x-1)!} = e^{-\lambda} \lambda \sum_{x \geq 1} \frac{\lambda^{x-1}}{(x-1)!}$$

And we see that, by the sum definition of the exponential function:

$$\sum_{x \geq 1} \frac{\lambda^{x-1}}{(x-1)!} = \sum_{x \geq 0} \frac{\lambda^x}{x!} = e^\lambda$$

This simplifies with the term $e^{-\lambda}$ above so that we obtain $E[X] = \lambda$.

Let's now compute the MLE of λ . The likelihood, by definition, is

$$L = \prod_{i=1}^m p(x_i; \lambda)$$

The log likelihood is:

$$\log L = \sum_{i=1}^m \log[p(x_i; \lambda)] = \sum_{i=1}^m -\lambda + x_i \log(\lambda) - \log(x_i!) = -m\lambda + \log(\lambda) \sum_{i=1}^m x_i - \sum_{i=1}^m \log(x_i!)$$

Taking the derivative with respect to λ of the log-likelihood we get

$$\partial_{\lambda} \log L = -m + \frac{1}{\lambda} \sum_{i=1}^m x_i$$

We obtain the critical point by setting the above derivative to 0.

$$\hat{\lambda} = \frac{\sum_{i=1}^m x_i}{m}$$

which is the sampled average. We want to check that this is a maximum of the log-likelihood (and so maximises the likelihood). Taking the second derivative of the log likelihood:

$$\partial_{\lambda}^2 \log L = -\frac{1}{\lambda^2} \sum_{i=1}^m x_i$$

we substitute $\hat{\lambda}$ in the above expression to get

$$\partial_{\lambda}^2 \log L(\hat{\lambda}) = -m^2 \frac{\sum_{i=1}^m x_i}{(\sum_{i=1}^m x_i)^2} = -m^2 \frac{1}{\sum_{i=1}^m x_i}$$

Since X is Poisson, the sum of the samples is strictly positive except when the samples are all null. Thus the second derivative of the log likelihood at the critical point is negative, and thus the critical point is a local maximum. In fact, since the second derivative is negative everywhere (apart from where the sample is all 0), the MLE is a global maximum.

When all the samples are 0, the log-likelihood is equal to $\log L = -m\lambda$: in this case it is easy to see that $\lambda = 0$ is the global maximum of the log-likelihood. (Note that the Poisson variable is defined for $\lambda > 0$)